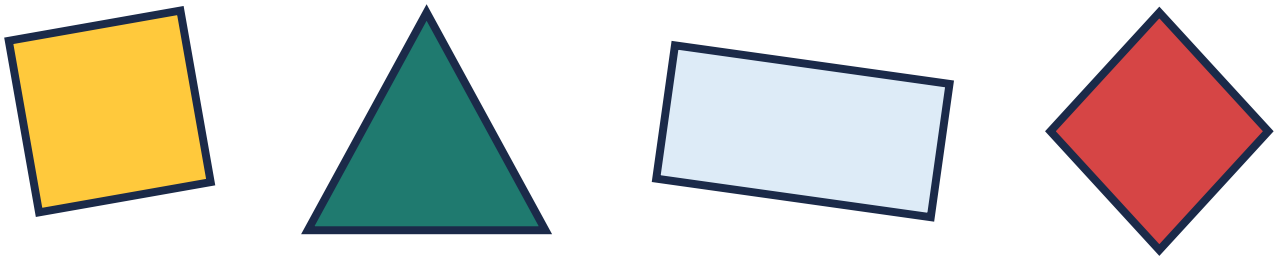
FOR PARENTS • LEVEL 4 HOLIDAY PROGRAM

Weekly Dose of Thinking Math

One dose a week for five weeks: short sets of questions for every Math Inquirers lesson, a tip for you, and the answers at the back.



Notice it. Test it. Explain why.

Math Inquirers Holiday Program
Preciouslyours Pte Ltd

BEFORE YOU START

00 FOR GROWN-UPS How to help

1. **Ask before you tell.** Start with "What do you notice?" and let your child answer first.
2. **Use real things.** Coins, notes, cups, string, paper strips or blocks. Try it with objects or a drawing before writing.
3. **Give hints in three steps.** Look again. Try it with objects. Try a smaller example. Say the answer last.
4. **Ask "How do you know?"** even when the answer is right. The reason matters as much as the answer.
5. **Praise the thinking.** "I like how you checked that" works better than "You are clever".
6. **Keep it short.** Ten minutes and two or three questions is enough. Stop while it is still fun.
7. **Answers are at the back.** Look only after your child has tried.

> **Hard question? Great. Getting stuck is where the thinking happens.**

One dose a week

Each week has two lessons and about 18 questions. Do two or three a day, Monday to Friday, and keep the weekend for the Go further questions if your child enjoys them.

The three levels

Start here is for everyone. **Think harder** stretches the idea. **Go further** and the **Investigation** are for when your child wants more, and it is fine to leave them for another day.

What your child is doing in class

Ten 90-minute live online lessons over five weeks, in small groups. Each lesson starts with a hands-on puzzle, then a short teaching part, then the workbook. These pages practise the same ideas at home.

W1 Factor Hunters

LESSON 1 • FACTORS AND MULTIPLES

Start here

1 Ali has 12 sweets. He puts them into bags so that every bag has the same number of sweets and none are left over. Use 12 counters, or draw 12 dots. Write down every number of sweets that one bag could have.

2 A frog starts at 0 and jumps along a number line in jumps of 7. Write the first five numbers it lands on.

3 Can 52 stickers be shared equally among 4 children with none left over? Try it: give each child 10 stickers first, then see how many are left to share.

4 Mei counts 3, 6, 9, 12 … (in 3s). Ravi counts 5, 10, 15, 20 … (in 5s). Write a number less than 20 that both of them say.

Think harder

5 How many factors does 72 have? List them in order.

6 One lighthouse flashes every 6 seconds. Another flashes every 9 seconds. They flash together now. After how many seconds will they next flash together?

What your child learns

Factors and multiples: listing factor pairs in order, common factors and common multiples, and square numbers.

Tip for you

Use 24 coins or buttons. How many different equal rows can your child make? Which numbers have the most rows?

Ask

"How do you know you have found every factor?"

7 Find all the common factors of 24 and 36.

Go further

8 How many numbers from 1 to 100 are multiples of 4 or multiples of 6 (or both)?

9 Mei has between 30 and 50 eggs. When she packs them in 4s, 1 egg is left. When she packs them in 5s, 1 egg is left. How many eggs does she have?

W1 Big Number Puzzles

LESSON 2 • NUMBERS TO 100 000

Start here

1 In 64 205, what is the digit 4 worth?

2 Write fifty-three thousand and eighty in numerals.

3 Mei's number is 27 649. Is it closer to 27 000 or to 28 000? Draw a number line from 27 000 to 28 000 and mark the halfway point (27 500) to help you.

4 What number is 1000 less than 70 230?

Think harder

5 I am a 5-digit number. All my digits are the same. Rounded to the nearest ten thousand, I am 60 000. Who am I?

6 Use the digit cards 1, 4, 6, 7 and 9 once each. Make the 5-digit number that is closest to 70 000.

7 Find the rule and write the next two numbers: 17 250, 19 500, 21 750, ____, ____

What your child learns

Numbers to 100 000: rounding, estimating, and riddles about 5-digit numbers.

Tip for you

Ask your child to round the price of a car or flat in the news to the nearest thousand, then to the nearest ten thousand.

Ask

"Which digit decides how you round?"

Go further

8 Ali counts on in 1000s, starting at 3456. He stops at the last number below 100 000. What is the last number he says? How many numbers does he say?

9 How many whole numbers round to 45 000 to the nearest thousand AND to 50 000 to the nearest ten thousand?

W2 LESSON 3 • FOUR OPERATIONS
Operation Detectives

Start here

1 Ali has 18 stickers. He buys 6 packs with 3 stickers in each pack. How many stickers does he have now? Then write ONE number sentence that shows what you did.

2 18 boys and 6 girls each buy 3 tickets. How many tickets are bought altogether? Then write ONE number sentence for it.

3 A ribbon 56 cm long is cut into 7 equal pieces. Ben takes 2 of the pieces. How long is Ben's ribbon?

4 At a camp there are 90 children. 30 of them make 5 equal teams for a game. After the game, one whole team goes home. How many children are left at the camp?

Think harder

5 There are 2 340 red and blue beads. There are 4 times as many blue beads as red beads. How many red beads are there?

6 Put ONE pair of brackets in $3 + 5 \times 6 - 2$ to make the answer 46.

What your child learns

Four operations: order of operations, brackets, missing signs and bar models.

Tip for you

Write $6 + 4 \times 3$ and ask for the answer. Then add brackets in different places and see how the answer changes.

Ask

"Which do you do first, and why?"

7 Write +, −, × or ÷ in each gap to make this true: 12 __ 4 __ 5 = 15

Go further

8 Use four 4s and any of +, −, ×, ÷ and brackets to make each of the numbers 1, 2, 3, 4 and 5. Which one was hardest?

9 Ali has $56 more than Ben. Together they have $300. How much money does Ben have?

W2 Fraction Frontier

LESSON 4 • FRACTIONS

Start here

1 Ravi eats 11 slices of pizza. Each pizza is cut into 3 equal slices. Draw circles cut into 3 slices to help you. How many whole pizzas did he eat, and how many slices of another pizza?

2 Mei has 2 whole chocolate bars and 3 more pieces. Each bar is broken into 4 equal pieces. Draw the bars. How many pieces does she have altogether?

3 There are 32 counters. Put them equally into 4 groups. How many counters are in one group? Ali takes one group. What fraction of the counters does he take?

4 Put 18 counters equally into 3 groups. Give 2 of the groups to Mei. How many counters does Mei get?

Think harder

5 Ben ate 7/4 pizzas and Ravi ate 1 5/8 pizzas. Who ate more?

6 Siti has 40 beads. 3/8 of them are blue. How many beads are NOT blue?

What your child learns

Fractions: improper fractions and mixed numbers, comparing fractions, and fractions of a quantity with bar models.

Tip for you

Share a pizza or cake into equal slices. Ask for the number of slices as an improper fraction and as a mixed number.

Ask

"Can you show it with a bar?"

7 Put these fractions in order, smallest first: 2/3, 1/2, 5/6, 7/12

Go further

8 I am a fraction equal to 3/4. My numerator and denominator add up to 28. Who am I?

9 Mei spent 1/3 of her money on a book and 1/4 of her money on a pen. She had $25 left. How much money did she have at first?

W3 LESSON 5 • DECIMALS
Decimal Detectives

Start here

1 Ali has 3 one-dollar coins, 4 ten-cent coins and 7 one-cent coins. How much money is that? Write the amount in dollars, using a decimal point.

2 Ali has $0.90 and Mei has $0.89. Who has more money? How much more? (Think in cents.)

3 Is 7.46 closer to 7.4 or to 7.5? Draw a number line from 7.4 to 7.5 and mark the halfway point 7.45 to help you.

4 Ali buys 3 pens at $1.25 each. How much does he pay? (Add $1.25 three times, or think of $1.25 as 125 cents.)

Think harder

5 Find ALL the digits that the letter A can be: 3.A5 < 3.42 (A is one digit)

6 A number with 2 decimal places rounds to 10 when rounded to the nearest whole number. What is the smallest it can be? The largest?

What your child learns

Decimals: ordering, rounding, and decimals in money and measures.

Tip for you

Look at a supermarket receipt. Order the prices from smallest to largest and round the total to the nearest dollar.

Ask

"What is each digit worth?"

7 What number is halfway between 2.6 and 2.7? What number is halfway between 2.6 and 2.61?

Go further

8 A number with 3 decimal places rounds to 2.46 when rounded to 2 decimal places. Its digits add up to 20. What is the number?

9 3 apples and 1 pear weigh 0.95 kg. 1 apple and 1 pear weigh 0.45 kg. All the apples weigh the same. What is the mass of the pear?

W3 Angle Turns

LESSON 6 • ANGLES

Start here

1 Stand up and face the front. Turn all the way round until you face the front again: that is a full turn, 360°. How many degrees is a half turn? How many degrees is a quarter turn?

2 The corner of a page is a right angle, 90°. How many right angles make a full turn? How many degrees is a turn of 3 right angles (3/4 of a turn)?

3 Ali faces East. He turns a quarter turn clockwise (the way the clock hands go). Which direction does he face now? Stand up and try it.

4 A compass has 8 directions: N, NE, E, SE, S, SW, W, NW. Each step to the next one is 45°. Mei faces N and turns 3 steps anticlockwise. Which direction does she face now? How many degrees did she turn?

Think harder

5 Siti faces SE. She turns 225° clockwise. Which direction does she face now?

What your child learns

Angles: measuring with a protractor, estimating, and turns on a compass or clock.

Tip for you

Ask which angle the hands of a clock make at 4 o'clock, then check with a protractor.

Ask

"Is it more or less than a right angle?"

6 What is the angle between the clock hands at 10 o'clock? At 2 o'clock? What do you notice?

7 How many degrees does the minute hand turn from 7:15 to 7:50?

Go further

8 What is the angle between the hands of a clock at 3:30?

9 Five rays come out of one point and make 4 equal angles side by side. The angle between the two outside rays is 120°. How many angles can you find altogether? How many of them are right angles?

W4 Area Architects II

LESSON 7 • AREA AND PERIMETER

Start here

1 Draw a rectangle 9 cm long and 7 cm wide on squared paper (1 cm squares). How many squares are inside it? That is its area in cm².

2 A rectangle is 15 cm long and 4 cm wide. Walk your finger once round the edge, adding the sides as you go. How long is the way round (the perimeter)?

3 Use 12 square tiles (each 1 cm by 1 cm). Make a rectangle using all of the tiles. Write its length and breadth. Can you make a different rectangle with the 12 tiles? Draw each one.

4 A rectangle has an area of 56 cm². Its length is 8 cm. Find its perimeter. (Draw 8 squares in a row and add rows until you have 56 squares.)

Think harder

5 List every rectangle with whole-cm sides and an area of 30 cm². Which one has the smallest perimeter?

What your child learns

Area and perimeter: figures made of rectangles, missing lengths, and same area with different perimeters.

Tip for you

Draw a room plan on squared paper. Ask for the area of the floor and the length of the skirting around it.

Ask

"Did you find all the missing lengths first?"

6 A rectangle has whole-cm sides and a perimeter of 20 cm. List all the possible rectangles and their areas. Which has the largest area?

7 A 3 cm by 3 cm square is cut from each of the four corners of a 12 cm by 8 cm rectangle. Find the area and the perimeter of the shape that is left.

Go further

8 A square is cut into two identical rectangles. Each rectangle has a perimeter of 24 cm. Find the area of the square.

9 A gap is cut into the middle of the top side of a 10 cm by 6 cm rectangle. The gap is 2 cm wide and 4 cm deep, with square corners. Find the area and the perimeter of the shape that is left.

W4 Symmetry and Nets

LESSON 8 • SYMMETRY AND NETS

Start here

1 Hold a box or a dice. How many faces, edges and corners does a cube have? Count them one by one.

2 Fold a square piece of paper so that the two halves match exactly. How many different ways can you fold it like this? (Folds from corner to corner count too.)

3 Now fold a rectangle that is not a square so that the two halves match exactly. How many different ways can you fold it?

4 Look at a dice. Opposite faces add up to 7. Which number is on the face opposite 2?

Think harder

5 A cuboid is 3 cm long, 2 cm wide and 1 cm high. Its net is drawn on 1 cm squared paper. How many squares does the net cover?

6 Ben puts a dice on the table. The top face shows 3 and the front face shows 5. What numbers are on the bottom face and on the back face?

What your child learns

Symmetry and nets: finishing symmetric figures, lines of symmetry, and nets that fold into a cube.

Tip for you

Cut out a cross of six squares and see if it folds into a cube. Try other arrangements.

Ask

"Which faces are opposite each other?"

7 A dice is on a table. Mei walks round it and can see the top and the four side faces. The numbers she can see add up to 19. What number is on the bottom face?

Go further

8 Two dice are glued together face to face, and the two glued faces show the same number. The numbers on all the faces you can still see add up to 36. What number is on the glued faces?

9 Show why a net of a cube can never have 5 squares in a row. What is the most squares in a row a net can have?

W5 Data Detectives

LESSON 9 • TABLES AND LINE GRAPHS

Start here

1 On a line graph, each line on the scale is worth 10. Draw the scale from 30 to 40. A point is halfway between the 30 line and the 40 line. What number does it show?

2 A stall sold 45 bowls of laksa on Monday and 60 bowls on Tuesday. By how many bowls did sales go up?

3 A table shows 32 visitors on Monday, 48 on Tuesday and 20 on Wednesday. How many visitors came altogether?

4 A line graph goes up from 8 am to 10 am, then down until 12 pm. Sketch what it looks like. At what time was the number the highest?

Think harder

5 A table shows that Class 4A has 28 boys and 6 more girls than boys. Class 4B has 63 pupils, with 5 more girls than boys. Which class has more girls?

6 On a line graph each line on the scale is worth 20. A point is 3 lines above the 40 line. What number does it show?

What your child learns

Tables and line graphs: reading trends, finding missing values, and spotting misleading scales.

Tip for you

Track the temperature or your child's steps each day for a week and draw a line graph together.

Ask

"What does the steepest part tell us?"

7 A shop sold 300 kites in 4 days. Day 1: 65, Day 2: 80, Day 4: 90. How many kites were sold on Day 3?

Go further

8 A shop's sales go up by the same amount every week. In week 2 it sold 45 toys and in week 6 it sold 85 toys. How many toys did it sell in week 1? In which week did it first sell more than 100 toys?

9 A stall sold 250 bowls of noodles from Monday to Friday. Monday 40 and Friday 50. Tuesday and Wednesday were the same, and Thursday was 10 more than Tuesday. Find the sales for Tuesday, Wednesday and Thursday.

W5 Challenge Day

LESSON 10 • MIXED HEURISTICS

Start here

1 Two next-door numbers (like 7 and 8) multiply to make 90. What are they? Try some pairs and write each try in a list.

2 A farm has chickens and rabbits. There are 8 heads and 22 legs. How many rabbits are there? Draw 8 animals with 2 legs each first, then see how many legs you still need to add.

3 I think of a number. I add 15, then double it. I get 70. What is my number? Start at 70 and undo each step.

4 Ali is 6th from the front and 6th from the back of a queue. How many people are in the queue? Draw the queue.

Think harder

5 There are 20 bicycles and tricycles with 47 wheels altogether. How many tricycles are there?

6 Two next-door even numbers (like 6 and 8) multiply to make 624. What are they?

What your child learns

Choosing a strategy: guess and check, supposition, working backwards and logic grids.

Tip for you

When your child is stuck, ask "Can you draw it?" or "Can you try a number?" before giving any other hint.

Ask

"How did you start? What would you try next time?"

7 Ravi spent $15 on a book. Then he spent half of the money left on a bag. He had $18 left. How much money did he have at first?

Go further

8 How many whole numbers from 1 to 100 have the digit 7 in them at least once?

9 A farm has 20 chickens and goats. The chickens have 4 more legs altogether than the goats. How many chickens are there?

LOOK AFTER YOUR CHILD HAS TRIED

ALL LESSONS

Answers

Week 1 • Lesson 1: Factor Hunters

1. 1, 2, 3, 4, 6 or 12 sweets in a bag.
2. 7, 14, 21, 28, 35
3. Yes. 40 stickers are given out first and 12 are left. 12 shared among 4 is 3 each, so each child gets 13 (4 × 13 = 52).
4. 15
5. 12 factors: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72.
6. 18 seconds (6, 12, 18 and 9, 18).
7. 1, 2, 3, 4, 6, 12
8. 33. Multiples of 4: 25. Multiples of 6: 16. Both (multiples of 12): 8, counted twice. 25 + 16 – 8 = 33.
9. 41. One less than the number of eggs is a common multiple of 4 and 5 (20, 40, 60), so 21, 41 or 61; only 41 is between 30 and 50.

Week 1 • Lesson 2: Big Number Puzzles

1. 4000 (4 thousands)
2. 53 080
3. 28 000 (27 649 is past the halfway point 27 500).
4. 69 230
5. 55 555 (66 666 rounds to 70 000; 44 444 rounds to 40 000).
6. 69 741 (259 away; the closest number above 70 000 is 71 469, which is 1469 away).
7. Add 2250 each time: 24 000, 26 250.
8. 99 456. He says 97 numbers (3456, 4456, ... 99 456).
9. 500. To the nearest thousand: 44 500 to 45 499. To the nearest ten thousand: 45 000 to 54 999. Both: 45 000 to 45 499.

Week 2 • Lesson 3: Operation Detectives

1. 36 stickers. 18 + 6 × 3 = 18 + 18 = 36.
2. 72 tickets. (18 + 6) × 3 = 24 × 3 = 72.
3. 16 cm (56 ÷ 7 = 8, and 8 × 2 = 16).
4. 84 children (a team has 30 ÷ 5 = 6 children; 90 – 6 = 84).
5. 468 (5 units = 2 340, 1 unit = 468).
6. (3 + 5) × 6 – 2 = 8 × 6 – 2 = 46.
7. 12 ÷ 4 × 5 = 15 (the only way).
8. For example 1 = (4 + 4) ÷ (4 + 4); 2 = 4 ÷ 4 + 4 ÷ 4; 3 = (4 + 4 + 4) ÷ 4; 4 = 4 + (4 – 4) × 4; 5 = (4 × 4 + 4) ÷ 4. Other ways are possible.
9. $122 (300 – 56 = 244, 244 ÷ 2 = 122; Ali has $178).

Week 2 • Lesson 4: Fraction Frontier

1. 3 whole pizzas and 2 slices of another pizza (11 thirds = 3 2/3 pizzas).
2. 11 pieces (2 × 4 = 8, and 3 more). In fractions: 11 quarters = 11/4 = 2 3/4 bars.
3. 8 counters in a group. Ali takes 1 group out of 4, which is 1/4 of the counters (1/4 of 32 = 8).
4. 12 counters (each group has 6; 2 groups = 12). That is 2/3 of 18.
5. Ben. 7/4 = 14/8 = 1 6/8, which is more than 1 5/8.
6. 25 (1 unit = 5, blue = 15, not blue = 5 units = 25).
7. 1/2, 7/12, 2/3, 5/6 (6/12, 7/12, 8/12, 10/12).
8. 12/16. 3/4 has 3 + 4 = 7 parts; 28 ÷ 7 = 4, so 3 × 4 = 12 and 4 × 4 = 16.
9. $60. Book 4/12, pen 3/12, so 5/12 is left. 5 units = $25, 1 unit = $5, 12 units = $60.

Week 3 • Lesson 5: Decimal Detectives

1. $3.47
2. Ali has more: 90¢ is 1¢ more than 89¢.
3. 7.5 (7.46 is just past the halfway point 7.45).
4. $3.75
5. 0, 1, 2 and 3 (3.45 is more than 3.42)
6. Smallest 9.50, largest 10.49
7. 2.65 and 2.605
8. 2.459 (it is from 2.455 to 2.464; 2.45_ needs 9 to make 20; 2.46_ would need 8, but 2.468 rounds to 2.47)
9. 0.2 kg (2 apples = 0.95 – 0.45 = 0.5 kg, 1 apple = 0.25 kg, pear = 0.45 – 0.25)

Week 3 • Lesson 6: Angle Turns

1. 180° for a half turn and 90° for a quarter turn.
2. 4 right angles make a full turn. Three right angles are 270°.
3. S
4. SW. She turned 3 × 45° = 135°.
5. N (225° is 5 steps of 45°: S, SW, W, NW, N)
6. 60° and 60°. Both are 2 spaces apart the shorter way.
7. 210° (7 jumps of 5 minutes, and each jump is one clock space of 30°)
8. 75° (the minute hand points to 6; the hour hand is halfway from 3 to 4, so it is 2 and a half spaces from the 6: 30° + 30° + 15°)
9. 10 angles (4 + 3 + 2 + 1). Each small angle is 30°, so the right angles are the 3-part angles: 2 of them.

Week 4 • Lesson 7: Area Architects II

1. 63 cm² (7 rows of 9 squares).
2. 38 cm (15 + 4 + 15 + 4).
3. 1 by 12, 2 by 6 and 3 by 4 (a 6 by 2 is the same rectangle turned round).
4. 30 cm (the breadth is 7 cm; 8 + 7 + 8 + 7 = 30).
5. 1 by 30 (62 cm), 2 by 15 (34 cm), 3 by 10 (26 cm), 5 by 6 (22 cm). Smallest: 5 cm by 6 cm, 22 cm.
6. 1 by 9 (9 cm²), 2 by 8 (16 cm²), 3 by 7 (21 cm²), 4 by 6 (24 cm²), 5 by 5 (25 cm²). Largest: the 5 cm by 5 cm square, 25 cm².
7. Area 96 – 4 × 9 = 60 cm². Perimeter 40 cm, the same as the rectangle: each cut takes away 3 + 3 and adds 3 + 3.
8. 64 cm². Each rectangle is one whole side and one half side all round, which is 3 sides of the square. 24 ÷ 3 = 8 cm, and 8 × 8 = 64.
9. Area 60 – 8 = 52 cm². Perimeter 32 + 4 + 4 = 40 cm: the two 4 cm sides inside the gap cannot be pushed out.

Week 4 • Lesson 8: Symmetry and Nets

1. 6 faces, 12 edges, 8 corners.
2. 4 ways (across the middle both ways, and along both diagonals).
3. 2 ways (across the middle both ways). Folding along a diagonal does not make the halves match.
4. 5
5. 22 squares: 2 faces of 3 by 2 (6 each), 2 faces of 3 by 1 (3 each) and 2 faces of 2 by 1 (2 each). 12 + 6 + 4 = 22.
6. Bottom 4 (3 + 4 = 7). Back 2 (5 + 2 = 7).
7. 2. All six faces add up to 1 + 2 + 3 + 4 + 5 + 6 = 21, and 21 – 19 = 2.
8. 3. Two dice have 21 + 21 = 42 altogether. The hidden faces add up to 42 – 36 = 6, and they show the same number, so each is 3.
9. With 5 in a row, four squares wrap round to make a ring of walls and the 5th lands on top of the 1st, so two squares share a face and no face is left for the other end. The most is 4 in a row (for example the cross-shaped net).

Week 5 • Lesson 9: Data Detectives

1. 35
2. 15 bowls
3. 100
4. 10 am
5. Neither: both have 34 girls. (4A: 28 + 6 = 34. 4B: 63 − 5 = 58, 58 ÷ 2 = 29 boys, 29 + 5 = 34 girls.)
6. 100 (40 + 20 + 20 + 20)
7. 65 (300 − 65 − 80 − 90)
8. Week 1: 35 (4 jumps make 40, so 10 a week). First more than 100: week 8 (105).
9. Tuesday 50, Wednesday 50, Thursday 60. (Tue + Wed + Thu = 160; take away the extra 10, 150 shared by 3 = 50.)

Week 5 • Lesson 10: Challenge Day

1. 9 and 10
2. 3 rabbits (and 5 chickens). Suppose all chickens: 16 legs, 6 short, each swap adds 2.
3. 20 (70 ÷ 2 = 35, 35 − 15 = 20)
4. 11 (5 + 1 + 5)
5. 7 tricycles (and 13 bicycles). All bicycles: 40 wheels, 7 short, each swap adds 1.
6. 24 and 26
7. $51 (18 × 2 = 36, 36 + 15 = 51)
8. 19 (7, 17, 27, 37, 47, 57, 67, 87, 97 and 70 to 79; 77 is counted once)
9. 14 chickens (28 legs) and 6 goats (24 legs).